

Symmetry-Resolved Tetrahedral Shell Complexes and Radial Block-Jacobi Spectra

Benjamin Walker Mayes

Abstract—We develop a finite, symmetry-resolved extension of the tetrahedral simplicial carrier used in companion work. The first result is an exact decomposition of the oriented filled-tetrahedron chain under the natural S_4 action: $C_0 = A_1 \oplus T_2$, $C_1 = T_2 \oplus T_1$, $C_2 = T_1 \oplus A_2$, and $C_3 = A_2$. The two-dimensional irrep E is absent from the linear chain but occurs in the traceless symmetric square $\text{Sym}_0^2(T_2) = E \oplus T_2$ and in the six-point edge/rectangle orbit. We then construct a pure S_4 -equivariant tetrahedral annulus with $f = (18, 76, 116, 56)$ and homology $(1, 0, 1, 0)$, fill its inner boundary to obtain a 3-ball with $f = (19, 80, 122, 60)$, and iterate the annulus to obtain a radial family $f(K_r) = (14r - 9, 70r - 60, 112r - 102, 56r - 52)$ with vertex representation $(3r - 1)A_1 \oplus (r - 1)E \oplus (3r - 2)T_2$. For a declared confining operator $H_{r,\beta} = L_0(K_r) + \beta D_r$, we derive an exact E ladder and, at $\beta = 1$, prove compact resolvent of the infinite operator and fixed-index convergence of the finite-shell spectra. The A_1 and T_2 sectors reduce to rank-one 3×3 block-Jacobi systems, hence to matrix Weyl functions and scalar Möbius continued fractions; exact variational brackets and explicit low-eigenvalue enclosures follow. Finally, we give two representation-theoretic no-go results: no nonzero S_4 -equivariant linear map $E \rightarrow T_2$ exists, and a context capable of linearly coupling these sectors must contain a three-dimensional T_1 or T_2 irrep. All statements are finite or operator-theoretic mathematics on declared carriers. No atomic, particle, spacetime, or physical-Hamiltonian identification is made.

Index Terms—simplicial complexes, tetrahedral symmetry, S_4 representation theory, combinatorial Laplacians, discrete exterior calculus, Jacobi-type operators, block-tridiagonal operators, Weyl functions, spectral convergence.

I. SCOPE AND CONTRIBUTION BOUNDARY

The purpose of this paper is to collect a set of post-companion mathematical results that enlarge the finite tetrahedral carrier while preserving a narrow claim boundary. The results concern finite S_4 representation theory, pure simplicial shell constructions, a declared radial operator family, and its infinite-volume spectral reduction. The basic chain conventions and skew Hodge–Dirac setup are those of the companion finite-carrier paper; the deterministic reconstruction problem is treated separately in the companion projective-receipt paper. Neither companion is logically required for the new shell triangulation once the tetrahedral vertex action is fixed.

Benjamin Walker Mayes is an independent researcher. ORCID: 0009-0002-5813-0724.

Paper III, Review edition 1.0, 20 September 2026. IEEE A4 typeset edition.

The ordinary background is standard. Finite-group characters and Schur’s lemma are used in the usual sense [1]. Simplicial boundary operators and combinatorial Laplacians follow the familiar discrete-exterior-calculus and simplicial-Laplacian setting [2]–[4]. The infinite radial reduction belongs to the general spectral theory of Jacobi-type difference operators and Weyl functions [5].

a) *What is not claimed.*: The shell index is not a principal quantum number; the declared $H_{r,\beta}$ is not asserted to be a physical Hamiltonian; E and T_2 are representation labels, not particle names; the radial depth is an internal combinatorial function, not a physical radius; and no energy, action, mass, charge, or time unit is supplied. Any later correspondence to a physical system would require an independently specified carrier map and empirical test.

II. REPRESENTATION THEORY OF THE FILLED TETRAHEDRON

A. Chain decomposition

Let $P \cong \mathbb{R}^4$ be the natural permutation representation of S_4 on the vertices of a tetrahedron. With the increasing-order orientation convention, the chain spaces identify with exterior powers,

$$C_0 \cong \Lambda^1 P, \quad C_1 \cong \Lambda^2 P, \quad C_2 \cong \Lambda^3 P, \quad C_3 \cong \Lambda^4 P.$$

We use the conventional irreducible labels A_1, A_2, E, T_1, T_2 for the five real irreducible representations of S_4 .

Theorem II.1 (Oriented-chain decomposition). *Under the natural S_4 action,*

$$\boxed{C_0 = A_1 \oplus T_2}, \quad \boxed{C_1 = T_2 \oplus T_1}, \\ \boxed{C_2 = T_1 \oplus A_2}, \quad \boxed{C_3 = A_2}.$$

Consequently,

$$\mathcal{C} = C_0 \oplus C_1 \oplus C_2 \oplus C_3 = A_1 \oplus 2A_2 \oplus 2T_1 \oplus 2T_2.$$

In particular, E is absent from the linear chain carrier.

Proof. The natural permutation representation decomposes as $P = A_1 \oplus T_2$. Taking exterior powers and reducing the corresponding characters gives the displayed sequence. Equivalently, the decomposition follows directly from the character table of S_4 and the character formula for exterior powers. The dimensions 4, 6, 4, 1 provide the independent dimension check. $\square$

Corollary II.2 (Symmetry typing of adjacent Hodge channels). *The equivariant boundary maps couple only*

common irreducible sectors. Thus the three adjacent-degree exchange channels are

$$\boxed{T_2 : C_0 \leftrightarrow C_1}, \quad \boxed{T_1 : C_1 \leftrightarrow C_2},$$

$$\boxed{A_2 : C_2 \leftrightarrow C_3}.$$

with an isolated harmonic A_1 vertex mode.

Proof. Equivariance of each boundary map with the vertex-permutation action and Schur's lemma imply that inequivalent irreducible sectors cannot couple. The dimensions in Theorem II.1 then identify the active channel in each adjacent pair. In particular, $\ker B_1 \cong T_1$ and the uniform vertex vector spans the A_1 harmonic sector. $\square$

B. Quadratic traceless parent

Proposition II.3 (Quadratic $E \oplus T_2$ sector). *For the standard three-dimensional T_2 representation,*

$$\text{Sym}^2(T_2) = A_1 \oplus E \oplus T_2,$$

so after removing the scalar trace,

$$\boxed{\text{Sym}_0^2(T_2) = E \oplus T_2}.$$

Proof. The symmetric-square character is $\chi_{\text{sym}^2}(g) = \frac{1}{2}(\chi(g)^2 + \chi(g^2))$. Reducing this character against the five irreducible characters of S_4 gives one copy each of A_1, E, T_2 . The A_1 summand is the trace. $\square$

No-go II.4 (No linear $E \rightarrow T_2$ intertwiner). *There is no nonzero S_4 -equivariant linear map $J : E \rightarrow T_2$:*

$$\boxed{\text{Hom}_{S_4}(E, T_2) = 0}.$$

Proof. The two representations are inequivalent irreducibles, so the statement is Schur's lemma. $\square$

The useful interpretation of Proposition II.3 and No-go II.4 is therefore representation-theoretic: E and T_2 can appear as sibling summands of a common quadratic parent, but they cannot be identified by an S_4 -equivariant linear isomorphism.

III. A REPEATED-SHELL GRAPH WARM-UP

Before constructing a pure three-dimensional shell complex, it is useful to record an exactly separable graph family. Let

$$G_r = P_r \square K_4,$$

where P_r is the path on r shell labels and the diagonal S_4 action acts on the K_4 factor.

Proposition III.1 (Repeated-shell decomposition). *The vertex carrier of G_r is*

$$\boxed{\mathcal{V}_r = rA_1 \oplus rT_2}.$$

Its graph Laplacian separates as

$$L(G_r) = L(P_r) \otimes I_4 + I_r \otimes L(K_4).$$

Writing

$$\rho_j = 2 - 2 \cos \frac{\pi j}{r}, \quad j = 0, \dots, r-1,$$

one has

$$\boxed{\lambda_{j,A_1} = \rho_j}, \quad \boxed{\lambda_{j,T_2} = \rho_j + 4}.$$

Proof. The representation statement follows from $\mathbb{R}^4 = A_1 \oplus T_2$ on every shell. The Laplacian identity is the standard Cartesian-product formula. Since $L(K_4)$ has eigenvalue 0 on A_1 and 4 on T_2 , the spectrum separates as stated. $\square$

This graph family supplies an exact radial-mode index within a declared graph product, but it is not itself a pure face-glued three-dimensional tetrahedral complex.

IV. A PURE S_4 -EQUIVARIANT TETRAHEDRAL ANNULUS

A. Construction

Take outer shell vertices O_i and inner shell vertices I_i , $i = 0, 1, 2, 3$. For each tetrahedral edge ij , introduce one rectangle-center vertex R_{ij} , and for each tetrahedral face introduce one prism-center vertex P_ℓ . Thus

$$4 + 4 + 6 + 4 = 18$$

vertices are used. Each of the four triangular prisms between corresponding inner and outer faces is triangulated by retaining the two triangular shell faces, splitting each side rectangle through R_{ij} , and coning each resulting boundary triangle to the prism center. The complete tetrahedron list is supplied in the reproduction package.

Theorem IV.1 (Pure annulus). *The resulting pure simplicial complex A has*

$$\boxed{f(A) = (18, 76, 116, 56)}.$$

Its boundary is exactly the disjoint union of the inner and outer tetrahedral 2-spheres, and its Betti numbers are

$$\boxed{(b_0, b_1, b_2, b_3) = (1, 0, 1, 0)}.$$

The full set of 56 tetrahedra is invariant under all 24 vertex permutations in S_4 .

Proof. Enumerating all nonempty faces of the 56 listed tetrahedra and deduplicating by vertex set gives the stated f -vector. Exactly eight triangles have tetrahedral incidence one: the four inner and four outer shell faces; every other triangular face has incidence two. Exact boundary-matrix ranks are

$$\text{rank } B_1 = 17, \quad \text{rank } B_2 = 59, \quad \text{rank } B_3 = 56,$$

which gives the displayed Betti numbers. Relabeling O_i, I_i, R_{ij}, P_ℓ by any $g \in S_4$ preserves the tetrahedron list. $\square$

B. Vertex representation and Laplacian blocks

The four vertex orbits have permutation representations

$$\begin{aligned} O &\cong A_1 \oplus T_2, & I &\cong A_1 \oplus T_2, \\ R &\cong A_1 \oplus E \oplus T_2, & P &\cong A_1 \oplus T_2. \end{aligned}$$

Therefore

$$C_0(A) = 4A_1 \oplus E \oplus 4T_2.$$

For the unit-weight vertex graph Laplacian, the multiplicity-space blocks in orbit order (O, I, R, P) are

$$L_{A_1}^A = \begin{pmatrix} 7 & -1 & -\sqrt{6} & -3 \\ -1 & 7 & -\sqrt{6} & -3 \\ -\sqrt{6} & -\sqrt{6} & 6 & -\sqrt{6} \\ -3 & -3 & -\sqrt{6} & 9 \end{pmatrix},$$

$$L_{T_2}^A = \begin{pmatrix} 11 & -1 & -\sqrt{2} & 1 \\ -1 & 11 & -\sqrt{2} & 1 \\ -\sqrt{2} & -\sqrt{2} & 6 & \sqrt{2} \\ 1 & 1 & \sqrt{2} & 9 \end{pmatrix},$$

while the single E copy has eigenvalue 6 with multiplicity two.

V. CENTER FILLING AND RADIAL REFINEMENT

A. Center-filled 3-ball

The annulus has an inner boundary and therefore does not itself contain a central core. Add one S_4 -fixed center C and cone it to the four inner-shell triangles.

Theorem V.1 (Center-filled shell). *The resulting pure 3-ball has*

$$f = (19, 80, 122, 60),$$

Betti numbers

$$(1, 0, 0, 0),$$

and vertex representation

$$5A_1 \oplus E \oplus 4T_2.$$

Its only boundary is the outer tetrahedral sphere.

Proof. Four center-to-inner-shell tetrahedra fill the inner boundary of Theorem IV.1. Exact ranks are 18, 62, 60 for B_1, B_2, B_3 , respectively, giving the displayed homology. Adding the fixed center adds one A_1 to the annulus vertex representation. $\square$

B. Arbitrary finite refinement

Let K_1 be the center-filled single tetrahedral shell, and define recursively

$$K_{r+1} = K_r \cup_{S_{\text{tet}}} A$$

by gluing a fresh copy of the annulus along the current outer tetrahedral boundary.

Theorem V.2 (Radial shell recurrence). *For every $r \geq 1$,*

$$f(K_r) = (14r - 9, 70r - 60, 112r - 102, 56r - 52).$$

Every finite K_r is a pure tetrahedral 3-ball, and

$$C_0(K_r) = (3r - 1)A_1 \oplus (r - 1)E \oplus (3r - 2)T_2.$$

Proof. Each annulus attachment adds 14 new vertices, 70 new edges, 112 new faces, and 56 new tetrahedra after subtracting the shared tetrahedral boundary. The topological assertion follows by gluing a 3-ball to $S^2 \times I$ along one boundary sphere. The new vertex orbits contribute $3A_1 \oplus E \oplus 3T_2$ at every step, giving the representation formula by induction. $\square$

The construction itself supplies an integer depth. We write

$$d(C) = 0, \quad d(S_s) = 2s - 1, \quad d(R_s) = d(P_s) = 2s.$$

This is a combinatorial depth function only.

VI. A DECLARED CONFINING OPERATOR FAMILY

The following operator is introduced as a mathematical family on the refined carrier; it is not derived as a physical Hamiltonian.

Definition VI.1 (Radial graph operator). For $\beta \geq 0$, let

$$H_{r,\beta} = L_0(K_r) + \beta D_r,$$

where L_0 is the unit-weight vertex Laplacian and D_r is multiplication by the depth d .

Because both terms are S_4 -invariant, $H_{r,\beta}$ splits into A_1, E, T_2 isotypic blocks.

Proposition VI.2 (Exact E ladder). *For every annulus layer $s = 1, \dots, r - 1$, the E doublet carried by the rectangle-center orbit is an exact eigenspace with*

$$\lambda_{E,s} = 6 + 2s\beta$$

of multiplicity two.

Proof. Only the rectangle-center orbit contains E . The E component has unit-Laplacian eigenvalue 6, the depth is constant $2s$ on that orbit, and there are no interlayer E copies on any other vertex orbit that could couple to it. $\square$

The remainder of the spectral analysis will specialize to the declared benchmark $\beta = 1$. This specialization is a mathematical normalization inside Definition VI.1; no empirical parameter is inferred from it.

VII. INFINITE REFINEMENT AND SPECTRAL CONVERGENCE

Let $K_\infty = \bigcup_{r \geq 1} K_r$ and define on $\ell^2(V(K_\infty))$

$$H_\infty = L_0(K_\infty) + D.$$

The graph has uniformly bounded degree, with maximum degree 17, so the graph Laplacian is bounded.

Theorem VII.1 (Compact resolvent). *The operator H_∞ has compact resolvent. Consequently its spectrum is purely discrete and may be written*

$$\lambda_1^\infty \leq \lambda_2^\infty \leq \dots \rightarrow \infty.$$

Proof. The multiplication operator D has finite multiplicity at each finite depth and tends to infinity along every sequence escaping to radial infinity. Therefore $(D+1)^{-1}$ is compact. Since L_0 is bounded, $H_\infty = D + L_0$ is a bounded perturbation of D , and the resolvent identity preserves compactness. $\square$

Let H_r denote the finite free-boundary operator on K_r . An outer-shell vertex has degree 10 in K_r and degree 17 after the next annulus is attached. If P_{S_r} projects onto the four outer-shell coordinates, then the principal compression of the infinite operator is

$$\boxed{H_r^D = P_r H_\infty P_r = H_r + 7P_{S_r}}.$$

Theorem VII.2 (Fixed-index spectral convergence). *For every fixed eigenvalue index k ,*

$$\boxed{\lambda_k(H_r) \longrightarrow \lambda_k(H_\infty)}.$$

More precisely, if $\mu_k(r) = \lambda_k(H_r^D)$ and $\nu_k(r) = \lambda_k(H_r)$, then

$$\mu_k(r) \downarrow \lambda_k(H_\infty)$$

and, for any r -independent upper bound C_k on the first k free-boundary eigenvalues for sufficiently large r ,

$$\boxed{0 \leq \mu_k(r) - \nu_k(r) \leq \frac{7C_k}{2r-1}}.$$

Proof. Nested finite-support subspaces form a form core for H_∞ , so the Dirichlet eigenvalues decrease to the infinite-volume eigenvalues by the min-max principle. On the first- k free-boundary spectral subspace, $H_r \geq D_r$ gives outer-shell mass at most $C_k/(2r-1)$. The boundary correction $7P_{S_r}$ therefore shifts the Rayleigh quotient by at most $7C_k/(2r-1)$. The two estimates squeeze $\nu_k(r)$ to the same limit. $\square$

VIII. BLOCK-JACOBI AND WEYL REDUCTION

The A_1 and T_2 multiplicity spaces may be grouped into radial cells

$$x_s = (S_s, R_s, P_s)^T.$$

At $\beta = 1$, the A_1 first cell is

$$A_1^{(A_1)} = \begin{pmatrix} 9 & -\sqrt{6} & -3 \\ -\sqrt{6} & 8 & -\sqrt{6} \\ -3 & -\sqrt{6} & 11 \end{pmatrix},$$

and for $s \geq 2$,

$$A_s^{(A_1)} = \begin{pmatrix} 2s+13 & -\sqrt{6} & -3 \\ -\sqrt{6} & 2s+6 & -\sqrt{6} \\ -3 & -\sqrt{6} & 2s+9 \end{pmatrix},$$

with intercell block

$$B_{A_1} = \begin{pmatrix} -1 & 0 & 0 \\ -\sqrt{6} & 0 & 0 \\ -3 & 0 & 0 \end{pmatrix}.$$

The central coordinate has diagonal 4 and coupling -2 to S_1 .

For T_2 ,

$$A_1^{(T_2)} = \begin{pmatrix} 13 & -\sqrt{2} & 1 \\ -\sqrt{2} & 8 & \sqrt{2} \\ 1 & \sqrt{2} & 11 \end{pmatrix},$$

$$A_s^{(T_2)} = \begin{pmatrix} 2s+17 & -\sqrt{2} & 1 \\ -\sqrt{2} & 2s+6 & \sqrt{2} \\ 1 & \sqrt{2} & 2s+9 \end{pmatrix}, \quad s \geq 2,$$

$$B_{T_2} = \begin{pmatrix} -1 & 0 & 0 \\ -\sqrt{2} & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}.$$

Both intercell blocks have rank one.

Theorem VIII.1 (Matrix continued fraction). *For $X \in \{A_1, T_2\}$, define the tail Schur matrix*

$$M_s^{(X)}(z) = zI_3 - A_s^{(X)} - B_X[M_{s+1}^{(X)}(z)]^{-1}B_X^T.$$

For fixed z off the spectrum, $[M_s^{(X)}(z)]^{-1} = O(s^{-1})$, so the infinite tail is represented by a convergent matrix continued fraction. Since $B_X = b_X e_1^T$ has rank one, the scalar Weyl component

$$m_s(z) = e_1^T [M_s(z)]^{-1} e_1$$

satisfies the exact Möbius recursion

$$\boxed{m_s = g_s + \frac{c_s^2 m_{s+1}}{1 - d_s m_{s+1}}},$$

where $g_s = e_1^T R_s^{-1} e_1$, $c_s = e_1^T R_s^{-1} b$, $d_s = b^T R_s^{-1} b$, and $R_s = zI - A_s$.

The A_1 limiting eigenvalues satisfy

$$\boxed{z - 4 - 4m_1^{(A_1)}(z) = 0},$$

while the T_2 limiting eigenvalues are the real poles of $m_1^{(T_2)}$.

A. Variational enclosures

The coupling norms are

$$\|B_{A_1}\| = 4, \quad \|B_{T_2}\| = 2.$$

For the constant parts of the radial cells,

$$\lambda_{\min}(C_{A_1}) = 9 - \sqrt{33},$$

while a sufficient exact lower bound in the T_2 sector is

$$\lambda_{\min}(C_{T_2}) \geq 6 - 2\sqrt{2}.$$

Cutting after cell N gives tail lower bounds

$$\boxed{\tau_{A_1}(N) = 2N - 1 - \sqrt{33}},$$

$$\boxed{\tau_{T_2}(N) \geq 2N + 2 - 2\sqrt{2}}.$$

Let H_N be the principal N -cell compression and $L_N = H_N - \|B\|Q_N$, with Q_N the projector onto the final cell. Once the tail floor lies above the spectral window of interest,

$$\boxed{\lambda_k(L_N) \leq \lambda_k(H_\infty) \leq \lambda_k(H_N)}.$$

Exact integer characteristic polynomials and rational Sturm isolation at $N = 8$ give

$$\begin{aligned} 2.066766567633982 &< \lambda_{A_1,1}^\infty < 2.066766567637642, \\ 4.191538100153076 &< \lambda_{A_1,2}^\infty < 4.191538100335074, \\ 6.686030497697500 &< \lambda_{T_2,1}^\infty < 6.686030497697504, \\ 8.967031711187236 &< \lambda_{T_2,2}^\infty < 8.967031711187948. \end{aligned}$$

The decimal intervals are certificate outputs from exact integer polynomials; they are not empirical measurements.

B. Asymptotic radial recurrence

Eliminating the local R_s, P_s coordinates gives a scalar three-term recurrence

$$u_{s+1} = Q_X(s, z)u_s - P_X(s, z)u_{s-1},$$

with rational coefficients. As $s \rightarrow \infty$,

$$\begin{aligned} P_{A_1} &\rightarrow 1, & Q_{A_1} &= 2s - z - 2 + O(s^{-1}), \\ P_{T_2} &\rightarrow 1, & Q_{T_2} &= 2s + 14 - z + O(s^{-1}). \end{aligned}$$

Thus the minimal tail is of discrete-Stark/Bessel type. The exact named object used here is the matrix Weyl function / continued fraction; no reduction of the full rational recurrence to an ordinary scalar Bessel or hypergeometric family is asserted.

IX. SIX-EDGE RELATION ALGEBRA AND SECOND-ORDER PAIR CURRENTS

The six unordered tetrahedral edges form a permutation module

$$\mathbb{R}^6 = A_1 \oplus E \oplus T_2.$$

There are exactly two nontrivial S_4 orbits of unordered pairs of distinct edge coordinates: adjacent edges and opposite edges.

Let A_{adj} and A_{opp} be the corresponding adjacency matrices, and let

$$L_{\text{adj}} = 4I - A_{\text{adj}}, \quad L_{\text{opp}} = I - A_{\text{opp}}.$$

Then

$$P_{A_1} = \frac{1}{6}J, \quad P_{T_2} = \frac{1}{2}L_{\text{opp}}, \quad P_E = I - P_{A_1} - P_{T_2},$$

and therefore

$$\boxed{L_{\text{opp}} = 2P_{T_2}}, \quad \boxed{L_{\text{adj}} = 6P_E + 4P_{T_2}}.$$

The two relation Laplacians order E and T_2 oppositely, so symmetry alone does not choose a preferred ordering or a unique positive linear combination.

No-go IX.1 (No source-forced universal relation ratio). *The finite hard gates and unary edge route weights of the companion source grammar do not determine a universal scalar ratio $w_{\text{opp}}/w_{\text{adj}}$ for the two relation Laplacians.*

The no-go is a type statement: a binary matching gate and a state-dependent weight on a single edge are not

second-order edge-pair couplings. Relation counts alone also fail to decide the ratio: equal weight per relation pair gives one convention, while equal total weight per relation orbit gives another.

A coefficient-free second-order object can nevertheless be derived from any declared nonnegative six-edge source lane λ .

Definition IX.2 (Quadratic pair-current lift). Set

$$\mathbb{J}_{ef}^{(2)} = (1 - \delta_{ef})\lambda_e\lambda_f.$$

The adjacent and opposite relation masks have exact source expressions

$$\boxed{A_{\text{adj}} = |B_1|^T |B_1| - 2I}, \quad \boxed{A_{\text{opp}} = P_\Theta^T P_\Theta - I},$$

where P_Θ is the 3×6 matching-address matrix. Therefore

$$J_{\text{adj}}^{(2)} = \frac{1}{2}\lambda^T A_{\text{adj}} \lambda, \quad J_{\text{opp}}^{(2)} = \frac{1}{2}\lambda^T A_{\text{opp}} \lambda.$$

If $W_p = \sum_{e \in M_p} \lambda_e$, then the same quantities are recovered from the existing matching aggregates:

$$\boxed{J_{\text{opp}}^{(2)} = \frac{1}{2} \left(\sum_p W_p^2 - \sum_e \lambda_e^2 \right)},$$

$$\boxed{J_{\text{adj}}^{(2)} = \frac{1}{2} \left[\left(\sum_e \lambda_e \right)^2 - \sum_p W_p^2 \right]}.$$

Their ratio is state dependent and is not promoted to a universal constitutive constant.

X. MINIMUM SYMMETRY TYPE FOR CROSS-SECTOR CONTEXT

The preceding sections are closed finite mathematics. A final representation-theoretic corollary identifies what symmetry type an auxiliary context would need in order to linearly couple E and T_2 .

Theorem X.1 (Minimum context dimension). *Let R be an irreducible S_4 context representation. A nonzero equivariant map*

$$R \otimes E \longrightarrow T_2$$

can exist only for $R = T_1$ or $R = T_2$. In particular, the minimum possible context dimension is three.

Proof. The exact tensor products are

$$E \otimes A_1 = E, \quad E \otimes A_2 = E,$$

$$E \otimes E = A_1 \oplus A_2 \oplus E,$$

$$E \otimes T_1 = T_1 \oplus T_2, \quad E \otimes T_2 = T_1 \oplus T_2.$$

Thus T_2 is absent for all one- and two-dimensional context irreps and present for both three-dimensional irreps. $\square$

The three-state weight-one and weight-two Hamming shells of the matching-bit carrier transform as $A_1 \oplus E$, so they do not by themselves meet the criterion in Theorem X.1. This is a representation-type no-go, not a statement about coupling strength.

XI. LIMITATIONS AND OPEN PROBLEMS

The exact statements above separate naturally from several choices that remain conditional.

- 1) The pure shell complexes and their S_4 decompositions are intrinsic to the declared triangulations. By contrast, the operator $H_{r,\beta} = L_0 + \beta D$ is a declared mathematical family, not a uniquely sourced physical law.
- 2) The specialization $\beta = 1$ fixes an internal unit for the spectral analysis. No physical scale is attached to it.
- 3) The six-edge relation algebra determines the available invariant projectors but does not determine a universal adjacent/opposite coupling ratio.
- 4) The pair-current lift is a derived quadratic correlation tensor; interpreting it as a physical interaction requires a separate constitutive map.
- 5) The context theorem fixes the minimum representation type for linear cross-sector coupling; it does not determine a microscopic response operator or continuous coupling amplitude.
- 6) None of the carrier labels A_1, A_2, E, T_1, T_2 are identified here with atomic orbitals, particles, or physical field components. Such comparisons may motivate future tests but are outside the theorem claims of this paper.

XII. REPRODUCTION PACKAGE

The accompanying package contains the complete annulus and center-filled tetrahedron lists, exact certificate data for the radial spectral brackets, the T_2 context Clebsch–Gordan matrices, and a verifier script. The verifier reconstructs the simplicial f -vectors and boundary ranks, checks the principal S_4 character decompositions, verifies the six-edge relation algebra, and replays the representation-theoretic context criterion. The spectral certificate JSON files preserve the exact characteristic-polynomial coefficient vectors used for the stated low-eigenvalue enclosures.

XIII. CLAIM TABLE

TABLE I
PAPER-SCOPED CLAIM TABLE

ID	Statement	Status	Depends on
P3.1	$\text{Sym}_0^2(T_2) = E \oplus T_2$.	Proposition	S_4 characters
P4.1	$G_r = P_r \square K_4$ has $rA_1 \oplus rT_2$ and separated Laplacian spectrum.	Proposition	Cartesian product
T5.1	Pure S_4 annulus: $f = (18, 76, 116, 56)$, $b = (1, 0, 1, 0)$.	Theorem	tetrahedron list
T6.1	Center-filled shell: $f = (19, 80, 122, 60)$, $b = (1, 0, 0, 0)$.	Theorem	T5.1
T6.2	K_r refinement and vertex-irrep recurrences.	Theorem	T5.1/T6.1
NG3.2	$\text{Hom}_{S_4}(E, T_2) = 0$.	No-go	Schur lemma
T8.1	H_∞ has compact resolvent and fixed-index finite-shell convergence.	Conditional theorem	declared $H_{r,1}$
NG10.1	Current finite source types do not fix a universal adjacent/opposite relation ratio.	No-go	relation types
P10.2	Exact quadratic pair-current masks and matching-aggregate formulas.	Proposition	B_1, P_Θ
T11.1	Minimum context irrep for $E \leftrightarrow T_2$ transfer has dimension 3.	Theorem	S_4 tensor products
T9.1	Rank-one block-Jacobi/Weyl reduction and spectral enclosures.	Conditional theorem	declared $H_{r,1}$

DATA AND CODE AVAILABILITY

A reproduction archive accompanies the source edition of this paper. Public archival deposition and a permanent identifier may be assigned separately. The package is intended to make the finite combinatorial and operator claims reproducible without relying on conversational context.

REFERENCES

- [1] J.-P. Serre, *Linear Representations of Finite Groups*, Graduate Texts in Mathematics 42, Springer, New York (1977). doi:10.1007/978-1-4684-9458-7.
- [2] A. N. Hirani, *Discrete Exterior Calculus*, PhD thesis, California Institute of Technology (2003). doi:10.7907/ZHY8-V329.
- [3] D. Horak and J. Jost, “Spectra of combinatorial Laplace operators on simplicial complexes,” *Advances in Mathematics* 244, 303–336 (2013). doi:10.1016/j.aim.2013.05.007.
- [4] O. Knill, “The Dirac operator of a graph,” arXiv:1306.2166 (2013).
- [5] G. Teschl, *Jacobi Operators and Completely Integrable Nonlinear Lattices*, Mathematical Surveys and Monographs 72, American Mathematical Society, Providence (2000). ISBN 0-8218-1940-2.
- [6] U. Pachner, “P.L. homeomorphic manifolds are equivalent by elementary shellings,” *European Journal of Combinatorics* 12, 129–145 (1991). doi:10.1016/S0195-6698(13)80080-7.
- [7] W. B. R. Lickorish, “Simplicial moves on complexes and manifolds,” *Geometry & Topology Monographs* 2, 299–320 (1999). arXiv:math/9911256.
- [8] B. W. Mayes, “Finite Tetrahedral Complexes and Skew Hodge–Dirac Support Flow,” Paper I, Review edition 1.0 (16 September 2026).
- [9] B. W. Mayes, “Projective Reconstruction from Simplex Stocks and Incidence Currents,” Paper II, Review edition 1.0 (16 September 2026).